

JEFFREY TSENG

<https://github.com/jefft72> | jtseng.org | [linkedin.com/in/j-tseng/](https://www.linkedin.com/in/j-tseng/)
Purdue University | tseng94@purdue.edu

PROFESSIONAL SUMMARY

AI engineer specializing in **multi-agent systems** and agentic infrastructure, with hands-on experience building **self-evolving agent harnesses, RAG pipelines, and production backend systems**.

EDUCATION

Purdue University, West Lafayette, IN, Bachelor of Science in Computer Science Expected Graduation: *May 2028*
Relevant Classes: Web App Programming, Programming in C, Object Oriented Programming, Discrete Math

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, JavaScript, Flutter, React Native, C, Rust

Libraries and Tools: React, LangChain, LLMs, Hugging Face, NumPy, SciPy, Matlab, Gazebo, ROS, Node, Tailwind

RELEVANT EXPERIENCE

Judi Health / Capital Rx

Remote

Backend Software Engineer

June 2026 – August 2026

- Developing backend systems for healthcare claims adjudication
- Built a **Rust-based linter** enforcing type-safe calls across the claims processing pipeline

Diffex (diffex.ai) – Luntraa Inc

San Francisco, CA

Co-Founder

April 2026 – Present

- Awarded **\$50000** through the **Daytona Grid Startup Initiative**
- Building a **self-learning, model-agnostic AI agent harness** that autonomously evolves through iterative cycles
- Achieved a jump from **0% to 20%** on SWE-Bench after a **single self-evolution cycle**

UPlate (Mobile Application)

Remote

CEO & Co-Founder

August 2025 – Present

- Managing over **150 active users** at any given time & **engineered a custom RAG pipeline** using **semantic chunking** to pre-process nutritional datasets, ensuring high-relevancy context retrieval for the **Google Gemini API**
- Developed from the ground up using **Flutter** to implement a **custom, responsive UI** featuring a stateful, multi-page onboarding flow to seamlessly **collect user biometrics**
- **Overseeing** a team of **5 engineers**, conducting interviews, and meeting with investors and clubs to secure partnerships

Google Developer Groups

Purdue University

Technical Teaching Assistant

August 2025 – May 2026

- Taught strategies to make **RAG backends**, like **semantic chunking, vector databases, and cosine similarity searches**
- Designed and deployed take-home projects for **300+ students teaching React, Google Cloud, Vertex AI, & Tensorflow**
- **Mentored 150+** developer role students, providing technical assistance on **Flutter, Firebase, and full-stack projects**

Hack the Future

Purdue University

Team Lead

October 2025 – May 2026

- **Led cross-functional development teams** of **10+** student engineers to design and deploy full-stack applications for local nonprofits, integrating **React, Node.js/FastAPI, and MongoDB**
- Conducted **code reviews** and managed teams with **Git workflow**, fostering a **collaborative development environment**
- Collaborated directly with nonprofit clients **to implement user-centric solutions** that fit their needs

PERSONAL PROJECTS

OpenCraft

- **2026 HackPrinceton Winner, Dedalus Track**
- Engineered a **multi-agent coordination backend** for a Minecraft task-execution system, using **Supabase** (Postgres + Realtime + pgvector) as a **shared state layer** for world objects, chat, jobs, agent status, and persistent memory.
- Built a **RAG pipeline** over **agent-directive Markdown** using OpenAI embeddings and Supabase vector search, enabling **role-aware retrieval** of operational policies separate from factual world-state queries.
- Designed and validated an **end-to-end integration harness** that simulated **multi-agent** job contention, directive retrieval, memory writes, world-state updates, and chat-based task handoff against **live backend services**.

References Available Upon Request